Cell Design Specification

Cell Design Specification — VBF-T8R1FLASH-DS-01

Case: t8_r1_flash | **Date**: 2026-08-25 | **Design**: V4 Thermal-tuned (final)

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020 base)	parameter set Chen2020
Nominal capacity	3.45 Ah (design 3.40 Ah, simulation-verified 1C)	params_r1_v4.json / cell/r1_v4_1c.json:capacity_ah
Voltage window	4.20 – 2.50 V	parameter set
Midpoint (plateau) voltage	3.909 V	cell/r1_v4_energy.json:midpoint_voltage_v
Electrode footprint	65 mm × 1580 mm (unwound electrode, area 0.1027 m^2)	parameter set Electrode height/width
Stack thickness (excl. casing)	135.3 µm	cell/r1_v4_energy.json:thickness_m
Shell/casing dimensions	Not provided (no shell-thickness parameter)	honest note
Electrolyte formulation	EC/EMC + $LiPF_6$ class; high-transport override σ=1.4 S/m, D=4.5e-10 m^2/s, t■=0.50 (estimate)	params_r1_v4.json; t■ marked estimate
Cation transference number	0.50 (estimate, high-transference formulation)	params_r1_v4.json

2. Electrode and separator

Layer	Material	Thickness	Porosity	Density	Source
Positive electrode	NMC811	51.4 µm	0.335	3262 kg/m^3	parameter set + params_r1_v4
Negative electrode	Graphite	57.9 µm	0.25	1657 kg/m^3	parameter set + params_r1_v4
Separator	Polyolefin	12.0 µm	0.55	397 kg/m^3	parameter set + params_r1_v4
Positive current collector	Al	8.0 µm	–	2700 kg/m^3	params_r1_v4
Negative current collector	Cu	6.0 µm	–	8960 kg/m^3	params_r1_v4
Positive particle radius	–	1.00 µm	–	–	params_r1_v4 (nano-NMC, literature high-power practice)
Negative particle radius	–	1.50 µm	–	–	params_r1_v4 (fine graphite)

N/P ratio: 0.77 (theoretical, literature stoich windows Δx_{pos}=0.75, Δx_{neg}=0.86; approximate — set does not expose stoich-limit keys; literature stoich windows dx_pos=0.75 (NMC811 0.99->0.24), dx_neg=0.86 (graphite->LiC_6); approximate, set does not expose stoich-limit keys). Discharge balance check: 1C capacity 3.45 Ah from $\Delta x_{pos} \approx 0.27 \rightarrow 0.85$, $\Delta x_{neg} \approx 0.901 \rightarrow 0.03$ (derived from initial concentrations + 1C capacity).

3. Process design parameters

Parameter	Value	Formula	Unit
Areal density, positive	111.5	thickness × (1–porosity) × density	g/m^2
Areal density, negative	72.0	thickness × (1–porosity) × density	g/m^2
Compaction density, positive	2.169	electrode density × (1–porosity) ÷ 1000	g/cm^3
Compaction density, negative	1.243	electrode density × (1–porosity) ÷ 1000	g/cm^3
Electrolyte fill amount	3.93 mL (≈4.72 g @1.2 g/cm^3 literature)	pore volume × electrolyte density × fill factor (1.0)	mL / g
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; actual production-line value requires tuning	–

4. Mass breakdown (contract caliber: layer stack, electrolyte/casing excluded)

Layer	kg/m^2	Mass (g/cell)	Source
Positive electrode	0.11152	11.45	cell/r1_v4_energy.json:layer_kg_m2
Negative electrode	0.07200	7.39	cell/r1_v4_energy.json:layer_kg_m2
Positive current collector	0.02160	2.22	cell/r1_v4_energy.json:layer_kg_m2
Negative current collector	0.05376	5.52	cell/r1_v4_energy.json:layer_kg_m2
Separator	0.00214	0.22	cell/r1_v4_energy.json:layer_kg_m2
Total (contract mass)		**26.8 g**	cell/r1_v4_energy.json:mass_kg

5. Performance verification (vs entry-0 criteria)

Metric	Value	Criterion	Verdict	Source
Energy density	473.0 Wh/kg	≥ 446.18	✓ PASS	cell/r1_v4_energy.json
5C capacity retention	99.0 %	≥ 90 %	✓ PASS	cell/r1_v4_retention.json
Cell mass	26.8 g	≤ 40 g	✓ PASS	cell/r1_v4_energy.json
4C-charge@45°C T_max	58.8 °C (331.98 K)	≤ 333.15 K	✓ PASS	cell/r1_v4_4c.json
Lithium plating (4C charge)	min anode potential 0.0247 V > 0	no plating	✓ PASS	cell/r1_v4_4c.json:anode_potential_v

6. Design notes

• Baseline Chen2020: ED 400.8 Wh/kg, 5C retention 8.7%, mass 43.5 g (r1 evaluate, fail). Root cause of 5C failure: cathode solid diffusion (D=4e-15 m^2/s, r=5.22 µm, τ≈6800 s).
• Fixes (params_r1_v4.json, rationale in log round 2/3 propose+evaluate): nano particles (pos 1.0 µm, neg 1.5 µm) → τ≈250 s; high-transport electrolyte (σ=1.4 S/m, D=4.5e-10, t■=0.5 estimate); thin collectors (Al 8 µm, Cu 6 µm) → mass 43.5→26.8 g, ED→473; electrodes thinned 32% (Pareto balance); separator porosity 0.55; pouch cooling area 0.0075 m^2 (flatter form factor).
• 4C thermal attribution: V3 337.3 K → V5 (transport only) 334.7 K → V4 (transport + cooling area) 332.0 K. Transport/particles ≈ 2.6 K, cooling area ≈ 2.8 K (log round 3).
• Contract caliber: electrolyte/casing excluded from mass (parameter set lacks electrolyte density); 5C/1C retention uses self-consistent nameplate capacity (nominal 3.40 Ah vs actual 3.45 Ah, +1.5%).